

The Three Operational Layers of AI in System and Chip Design

Distinguishing point-task assistance and local subsystem optimization from cross-boundary system orchestration

Increasing Cross-Layer Scope & Multi-Tool Verification

LAYER 3 AI-Native System Design — Cross-Boundary Closed-Loop Orchestration (Architecture 2.0 Moonshot)

Operational Scope

- Cross-boundary closed-loop multi-tool search
- Workload ↔ Compiler ↔ RTL ↔ Physical ↔ Formal
- Co-explores microarchitecture & PDK signoff

Key Technical Mechanisms

- Multi-level IRs (CIRCT / MLIR dialect lowering)
- Rejectable multi-fidelity check harnesses
- Cryptographic lineage & design-loop cards

Governance & Signoff Boundary

- In-Order Human Commitment Authority
- Multi-property formal & physical signoff gates
- Governs daily code churn & tool dispersion

LAYER 2 AI-Driven Subsystem Optimization — Autonomous Deep Local Search Inside a Fixed Container

Operational Scope

- Autonomous search in a single subsystem
- Macro floorplanning & kernel autotuning
- DRAM controller timing & DSE sweeps

Key Technical Mechanisms

- Reinforcement learning (e.g. AlphaChip)
- Bayesian optimization & conformal models
- Domain-specific surrogate cost models

Governance & Signoff Boundary

- Bounding container frozen by architect
- Cannot alter cross-stack interfaces
- Bounded by Waterbed Effect & Amdahl limits

LAYER 1 AI-Assisted Engineering — Human-in-the-Loop Point Task Speedup Within Existing Workflows

Operational Scope

- Accelerates isolated drafting & typing
- RTL module generation & testbench templates
- Timing log parsing & script generation

Key Technical Mechanisms

- Open-loop foundation models & copilots
- Unstructured natural language & tokens
- Syntax-level completion & code search

Governance & Signoff Boundary

- Human acts as manual integration bus
- Generates syntax without signoff proof
- Physical qualification remains necessary

The Architecture 2.0 Ambition: Extend point assistance and subsystem search to cross-boundary exploration. Layer 3 aims to carry compact intent through real tools to physically realizable silicon; the moonshot remains open.